

Security+ 2.4 · Other Malware Types

Master CyberSec — Official CompTIA Security+ (SY0-701) Comprehensive Study Guide · **Lesson 2.4 — Other Malware Types (7:32)** · [Official YouTube Video ↗](#) · [Professor Messer Study Hub ↗](#)

Executive Lesson Summary: Keyloggers record keystrokes to steal cleartext passwords. Logic bombs trigger on a specific event (date, termination, keystroke) and remain dormant until activated. Rootkits hide in the OS kernel, making them nearly invisible to standard antivirus; detection requires booting from known-good media or reinstalling the OS. Trojans masquerade as legitimate software. Backdoors are covert access points maintained by attackers.

High-Yield Core Concepts Tested on Exam

- Rootkit hides in the OS kernel: nearly invisible (detect by booting from known-good media)
- Keylogger records all keystrokes in cleartext — steals passwords during transmission
- Logic bomb triggers on a specific EVENT (date/termination/action)
- Trojan disguises itself as legitimate software; backdoor provides persistent attacker access

Exam Tips & Classic Scenarios

- 'malware that hides itself within the OS kernel' = rootkit
- 'dormant until triggered by a specific event' = logic bomb
- 'records keystrokes to capture credentials' = keylogger
- 'legitimate-looking program that installs a backdoor' = trojan

Typical Exam Scenario: Scenario: 'A terminated employee had installed malware that detonates on a specific date' = logic bomb (insider threat).

Beyond the Video: Official CompTIA Exam Concepts, Protocols & Traps (Lesson 2.4 — Other Malware Types)

Why is this advanced coverage required for SY0-701? Video overviews provide foundational familiarity, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

- **WORM:** Write Once, Read Many: Immutable optical/cloud storage paradigm preventing data alteration or deletion during retention periods (critical for audit logs).

Classic CompTIA Traps & Question Framing:

- Avoid rote memorization: CompTIA exam prompts regularly adjust technical constraints requiring you to defend why one valid security control is the *MOST appropriate* or *FIRST step*.
- Scrutinize question qualifiers: 'without additional licensing costs' favors native OS tools; 'granular administrative command accounting' favors TACACS+ over RADIUS; 'non-repudiation' mandates asymmetric private key signing.

Practice Questions with In-Depth Solutions (10 Questions)

Detailed answer key with step-by-step technical rationale and comprehensive distractor breakdown provided below. Attempt answering independently first.

1. Malware that encrypts your files and demands payment to return them is:

- A)** Spyware
- B)** Ransomware
- C)** Rootkit
- D)** Worm

2. Which type of malware spreads automatically across networks without the need for a host file or user interaction?

- A)** Worm
- B)** Trojan
- C)** Spyware
- D)** Adware

3. Corporate scenario (Other Malware Types): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 2.4), an analyst evaluates the following technical requirement: 'Rootkit hides in the system KERNEL: almost invisible (boot from known-good media to detect)'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: Rootkit hides in the system KERNEL: almost invisible (boot from known-good...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

4. Corporate scenario (Other Malware Types): During a security review and compliance audit aligned with the CompTIA Security+ (SY0-701, Section 2.4) exam, an analyst evaluates the following technical requirement: 'Keylogger: records everything in cleartext — steals password on submission'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: Keylogger: records everything in cleartext — steals password on submission.
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

5. Corporate scenario (Other Malware Types): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 2.4), an analyst evaluates the following technical requirement: 'Logic bomb: detonated by EVENT (date/termination/action)'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: Logic bomb: detonated by EVENT (date/termination/action)...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

6. Corporate scenario (Other Malware Types): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 2.4), an analyst evaluates the following technical requirement: 'malware that hides itself within the OS kernel' = rootkit'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: 'malware that hides itself within the OS kernel' = rootkit.
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

7. Corporate scenario (Other Malware Types): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 2.4), an analyst evaluates the following technical requirement: 'dormant until triggered by a specific event' = logic bomb'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: 'dormant until triggered by a specific event' = logic bomb...

8. Corporate scenario (Other Malware Types): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 2.4), an analyst evaluates the following technical requirement: 'Scenario: a terminated employee has malware that he himself installed and that detonates on date X' = logic bomb (insider threat). What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to speed up system throughput
- B)** Implement the corresponding solution: Scenario: a terminated employee has malware that he himself installed and that det...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

9. Corporate scenario (Other Malware Types): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 2.4), an analyst evaluates the following technical requirement: 'Rootkit hides in the system KERNEL: nearly invisible (boot from known-good media to detect)'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: Rootkit hides in the system KERNEL: nearly invisible (boot from known-good...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

10. Corporate scenario (Other Malware Types): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 2.4), an analyst evaluates the following technical requirement: 'Keylogger: records everything in cleartext — steals password on submission'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: Keylogger: records everything in cleartext — steals password on submission...
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COMPREHENSIVE ANSWER KEY & DISTRACTOR BREAKDOWN

1) B — Ransomware.

✓ **Why it is correct:** The correct answer is 'Ransomware'. Ransomware is a type of malware that encrypts the victim's files and demands a ransom payment (usually in cryptocurrency) to provide the decryption key. The statement describes exactly this behavior: malware that encrypts files and demands payment to return them. CompTIA classifies ransomware as a critical threat, often propagated through phishing, vulnerability exploitation, or remote desktop services. Therefore, the option 'Ransomware' is the precise definition of the described attack.

✗ **Why the other options are wrong:**

- *Spyware*: Spyware is malicious software designed to collect information from the infected system without the user's consent, such as browsing habits, credentials, banking data, etc. It does not encrypt files or demand ransom. Its primary function is to spy and exfiltrate data. In the scenario, the goal is to extort money through file encryption, which is not a characteristic of spyware. Therefore, spyware does not apply.
- *Rootkit*: A rootkit is a set of malicious tools that allows an attacker to gain privileged access (root) and maintain persistence on the system, hiding its presence from detection mechanisms. Although it can be used in conjunction with ransomware, a rootkit itself does not encrypt files or demand payment. Its purpose is stealth, not extortion. The statement describes a financial extortion attack via encryption, not the installation of a rootkit. Thus, rootkit is not the correct answer.
- *Worm*: A worm is self-replicating malware that spreads automatically across the network, exploiting vulnerabilities, without necessarily needing a host file or user interaction. Although worms can carry payloads like ransomware, they are not defined by encrypting files and demanding ransom. The behavior described in the statement is specific to ransomware, not a worm. Therefore, worm is an incorrect classification for the attack in question.

2) A — Worm.

✓ **Why it's correct:** A Worm is a type of standalone malware that self-replicates and spreads automatically across networks, exploiting vulnerabilities in services or operating systems, without the need for a host file or user interaction. Unlike viruses, it does not need a carrier program to propagate. In CompTIA Security+ (domain 2.4 - 'Given a scenario, analyze indicators of malicious activity'), a worm is classically defined as malware that spreads on its own across the network, consuming bandwidth and compromising systems at scale. This is exactly what the question describes.

✗ **Why the others are wrong:**

- *Trojan*: A Trojan is malware that disguises itself as legitimate software to trick the user into executing it. It depends on user interaction to infect the system and does not self-replicate automatically across the network. Therefore, it does not meet the criterion of 'spreading automatically without user interaction'.
- *Spyware*: Spyware is malicious software designed to collect user information without consent, such as browsing habits, credentials, and personal data. It does not spread automatically across networks; it is typically installed through social engineering or bundled with other software. Thus, it is not the correct answer.
- *Adware*: Adware is software that displays unwanted advertisements and may redirect the browser, but it does not self-replicate or spread automatically across networks. Its dissemination depends on user installation or other malware. Therefore, it does not meet the question's criteria.

3) B — Implement the corresponding solution: Rootkit hides in the system KERNEL: almost invisible (boot from known-good....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 2.4 (Other Malware Types): Rootkit hides in the system KERNEL: almost invisible (boot from known-good media to detect). A rootkit is malware that embeds itself in the kernel or other privileged areas of the OS, making it extremely difficult to detect. Booting from known-good media (e.g., a clean USB or CD) allows scanning the system without the rootkit active, aiding detection and removal.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting principle and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

4) B — Implement the corresponding solution: Keylogger: records everything in cleartext — steals password on submission..

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 2.4 (Other Malware Types): Keylogger: records everything in cleartext — steals password on submission. Keyloggers capture keystrokes, including credentials, which can be mitigated by using multi-factor authentication, virtual keyboards, and endpoint protection.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

5) A — Implement the corresponding solution: Logic bomb: detonated by EVENT (date/termination/action)....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 2.4 (Other Malware Types): Logic bomb: detonated by EVENT (date/termination/action). A logic bomb is a type of malware that triggers when a specific condition is met, such as a date, a user action, or an employee termination.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks,

injections, or malware at runtime.

- *Allow shared access among administrators with generic credentials*: Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

6) B — Implement the corresponding solution: 'malware that hides itself within the OS kernel' = rootkit..

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 2.4 (Other Malware Types): 'malware that hides itself within the OS kernel' = rootkit. Rootkits operate at the kernel level to conceal their presence and that of other malware, making them extremely difficult to detect and remove. Mitigations include secure boot, integrity checks, and kernel-level monitoring.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials*: Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

7) D — Implement the corresponding solution: 'dormant until triggered by a specific event' = logic bomb....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 2.4 (Other Malware Types): 'dormant until triggered by a specific event' = logic bomb. A logic bomb is malicious code that executes when a specific condition is met.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detect incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

8) B — Implement the corresponding solution: Scenario: a terminated employee has malware that he himself installed and that det....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 2.4 (Other Malware Types): Scenario: a terminated employee has malware that he himself installed and that detonates on date X = logic bomb (insider threat). A logic bomb is a piece of code that triggers a malicious action when specific conditions are met (e.g., a date). The appropriate control is to implement the corresponding solution, which includes user behavior monitoring, least privilege, separation of duties, and proper offboarding procedures to mitigate insider threats.

✗ Why the others are wrong:

- *Temporarily disable audit logs to speed up system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or malware at runtime.
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9) A — Implement the corresponding solution: Rootkit hides in the system KERNEL: nearly invisible (boot from known-good....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 2.4 (Other Malware Types): Rootkit hides in the system KERNEL: nearly invisible (boot from known-good media to detect).

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
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10) B — Implement the corresponding solution: Keylogger: records everything in cleartext — steals password on submission....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 2.4 (Other Malware Types): Keylogger: records everything in cleartext — steals password on submission. A keylogger captures keystrokes in cleartext, allowing attackers to steal credentials as they are typed, including passwords before encryption.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

